

Neuronales Netz für den kalifornischen Hauspreis-Datensatz

Florian Adamczyk^a, Felix Hollfoth^b, Kolja Scriba^c, Björn Becker^d and Felix Neumann^e

^aTechnische Hochschule Mittelhessen, Matrikelnr. 8105234

^bAffiliation Two

^cAffiliation Three

^dAffiliation Three

^eJustus Liebig Universität, Matrikelnr. 8116639

This manuscript was compiled on March 26, 2026

Abstract—Dieser Ergebnisbericht dokumentiert die Ergebnisse der Gruppe 308 der Projektarbeit zu Maschinellem Lernen im Modul Künstliche Intelligenz 1. Die Aufgabenstellung lautet: Betrachten Sie den Datensatz zu den kalifornischen Hauspreisen, wie in der Vorlesung betrachtet. Entwickeln Sie ein neuronales Netz, um die Hauspreise zu beschreiben. Ordnen Sie Ihr Modell im Vergleich zu den in der Vorlesung behandelten Linearen Regression ein. Die Erarbeitung betrug pro Gruppenmitglied mindestens 75 Stunden und wurde in mehreren Meetings zwischen den Personen aufgeteilt. Neben der linearen Regression wurden auch andere Regressionsmodelle als Vergleichsgrundlage für das Neuronale Netz herangezogen. Der Schwerpunkt liegt allerdings auf der sukzessiven Verbesserung des Neuronalen Netzes, unter Anwendung der Vorlesungsinhalte. Die Optimierung der Regression erfolgt unter ständiger Kontrolle der Fehlergrößen wie dem Mean Squared Error (=MSE) und dem Root Mean Square Error (=RMSE), sowie dem coefficient of determination ($= R^2$) als quantitativer Vergleich der Modelle.

Keywords—neurales Netz, California Housing, TensorFlow, Regression, Machine Learning

1. EXPLORATIVE DATENANALYSE UND DATENBEREINIGUNG

Die explorative Datenanalyse dient der Aufnahme von Eckdaten des Datensatzes und dem Erlangen eines Gefühls für potenzielle Besonderheiten, welche in der Datenbereinigung für ein besseres Ergebnis sinnvoll behoben werden.

1.1. Analyse

Die Explorative Datenanalyse ergab, dass zwei der acht Features des Datensatzes cut-off-Werte besitzen. Eine unrealistisch hohe Anzahl an Werten befindet sich hierbei genau an der Grenze der aufgetragenen Werte. Zudem ist eine hohe Streuung der Werte mit einigen vermeintlichen Ausreißern direkt zu erkennen.

1.2. Bereinigung

Die Cut-off Werte der Features 'MedHouseVal' und 'HouseAge' wurden mit den Outliern über dem 98%-Quantil bei den Features 'AveRooms', 'AveBedrms', 'Population' und 'AveOccup' entfernt, da diese Artefakte darstellen und das Rauschen der Daten erhöhen.

1.3. Bereinigte Daten

Fig. 2 shows a 3D surface plot of the hyperbolic paraboloid $z = x^2 - y^2$.

This document class was prepared on Overleaf and compiled with pdf_{La}T_EX. No errors were found during the compilation process. Tau-class supports external editors, though additional setup may be required.

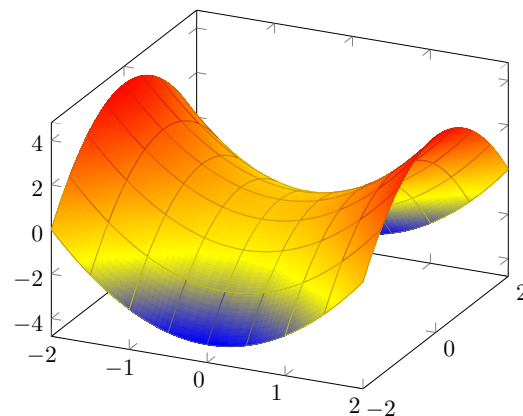


Figure 1. Hyperbolic paraboloid obtained from PGFPlots [3].

2. DOCUMENT STYLING

2.1. Document Type

Before the main title, a brief descriptor indicates the type of document (e.g., Lab Report, Research Article, etc). This is controlled by the `\doctype{}` command.

If no such label is needed, omit the command and the title will automatically reposition itself to maintain proper vertical spacing and visual alignment.

2.2. Abstract and Keywords

The abstract and keywords are declared in the preamble of the document, before the beginning of the document, and are automatically positioned during the title creation process.

2.3. Lettrine

The `\taustart{}` command, provides a personalized lettrine for the beginning of the first paragraph as shown in this example.

2.4. Table of Contents

This class includes a customized design for the table of contents, which is disabled by default. To enable it, insert the `\tableofcontents` command before your first section.

3. CROSS-REFERENCE COMMANDS*

Tau-class provides shorthand commands for cross-referencing figures, tables, code listings, and equations. To customize the label format, edit the `taubabel.sty` file. Label translations for Spanish are handled automatically.

- `\figref{}` — Uses the label style defined by `\figlabel`.
- `\figsref{}` — Uses the label style defined by `\figslabel`.
- `\tabref{}` — Uses the label style defined by `\tablalabel`.
- `\coderef{}` — Uses the label style defined by `\captionlabelcode`.
- `\equref{}` — Uses the label style defined by `\eqlabel`.

These commands generate fully hyperlinked references, both the label and the number, ensuring navigation to the referenced element when clicked in the PDF.

4. FIGURES AND TABLES

4.1. Figures

Fig. 2 shows a 3D surface plot of the hyperbolic paraboloid $z = x^2 - y^2$.

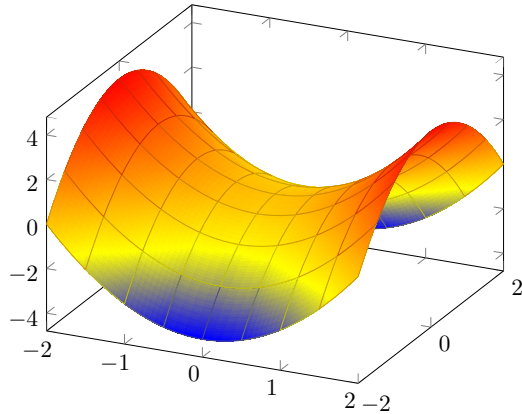


Figure 2. Hyperbolic paraboloid obtained from PGFPlots [3].

4.2. Tables

Table 1 shows an example table of some astronomical objects.

Table 1. Astronomical Object Data

Object	Type	Distance (Light Years)
Alpha Centauri	Star system	4.37
Betelgeuse	Red supergiant star	642.5
Andromeda	Spiral galaxy	2.537 million
Earth	Planet	0
Sirius	Binary star system	8.6

Note: The table contains data of some famous celestial objects.

The `\tabletext{}` command is used to add notes to tables easily.

5. DOCUMENT DATA

5.1. Front-Matter Footer Block*

The `\docinfo{}` command inserts a dedicated block at the bottom of the first column. Unlike floating elements, this block stays fixed in place — positioned visually like a footnote, although technically it is not one.

The `\docinfo{}` block was created to give you greater control over supplementary document data, such as publication dates, licensing information, extended author details, and other front-matter notes. You can fully customize its content to suit your needs. If it's not required, omit the command to remove this block entirely.

Note

This feature has been adapted to work seamlessly alongside the standard `\thanks{}` command, which is commonly used to indicate corresponding authors, equal contributions, or other acknowledgments.

If `\thanks{}` is not used, `\docinfo{}` will automatically adjust its position. The implementation was designed to handle any combination of front-matter elements you might need when starting your document. In this example template, you'll find a minimal demonstration that combines both to illustrate their interaction.

5.2. Headers and Footers

5.2.1. Headers

On all pages except the first, the document title appear in the header. Since twoside layout is enabled, its position alternates depending on whether the page is odd or even.

5.2.2. Footers

On odd-numbered pages, five elements display supplementary information for your document:

- `\thepage` — Including the total number of pages (only for the first page).
- `\footinfo{}` — For a short title or custom note.
- `\theday{}` — Displays the current date.
- `\organization{}` — For a university, institute, company, or any institutional affiliation.
- `\leadauthor{}` — For the main author name et al.

On even-numbered pages, the page number appears alongside the content of `\footinfo{}` and, on the other side, the `\leadauthor{}`.

In contrast to the header, the footer elements do not alternate their positions. The footer is explicitly configured for either odd or even pages, resulting in a fixed layout.

In all cases, you can freely rearrange or customize the order of these elements by modifying the class file to suit your needs. If any of them are not required, simply omit the corresponding command — the remaining elements will automatically adjust their spacing to remain evenly distributed.

6. TAU CUSTOM PACKAGES

6.1. Taubabel.sty

This package have all the commands that automatically translate from English to Spanish when this custom package is defined.

By default, this document has its content in English. However, at the beginning of the document you will find a recommendation when writing in Spanish.

You may modify this package if you want to use other language than English or Spanish. This will make easier to translate your document without having to modify the class document.

6.2. Tauenvs.sty

This package provides custom environments designed to enhance the visual presentation and structure of your document. Key examples include `tauenv`, `info`, and `note`.

Their style is defined by `tauenvstyle` — allowing you to customize their appearance according to your document's requirements.

An example using the `tauenv` environment is shown below.

Custom Title

This is an example of the custom title environment. To add a title type this command `[frametitle=Custom Title]` next to the beginning of this environment (as shown in this example).

The `info` and `note` are environments which have a predefined title and translate their title to Spanish automatically when this language is defined.

7. EQUATIONS

Equation 1 shows the Schrödinger equation as an example.

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \left[\frac{-\hbar^2}{2m} \nabla^2 + V(\mathbf{r}, t) \right] \Psi(\mathbf{r}, t) \quad (1)$$

The `amssymb` and `amsmath` packages were not required, as `STIX2` font incorporates mathematical symbols for writing quality equations.

Note

If you would like to change the values that adjust the spacing above and below the equations, change the `\eqskip` value until the preferred spacing is set. The default value is set to 9pt.

8. CODES

8.1. Coding with Minted*

The `minted` package offers customized features for adding codes. In addition, the template is designed to work seamlessly with Fira Code, a monospaced font specially crafted for programming environments.

If you are using a desktop app as `TeXstudio`, try these steps to make this package work on Windows:

1. Install Python — A stable version (e.g., v.3.11)
2. Open the terminal and type `pip install Pygments`.
3. Update if a newer version is available.
4. Go to `TeXstudio` settings and change the default compiler to `pdflatex -shell-escape -interaction=nonstopmode \%.tex`.
5. Compile and wait for the result.

Caution

Ensure that `pygments` is properly installed and added to your system's `PATH`. Otherwise, you may encounter compilation errors. Additionally, enable shell escape when compiling, as it is required for `minted` to process and highlight code.

In Overleaf, this package is easier to use since it does not require any additional installations or modifications — just add the code as shown in the example. [Code 1](#) shows a Python example.

```
# Fira Code Demo - Python
def main():
    # Common operators
    a = 10
    b = 20
    if a < b and a != 0:
        print("a is less than b")

    # Comparisons
    x = (a >= 5) and (b <= 30)

    # Walrus operator
    if (value := get_data()) is not None:
        process(value)

    # Lambda with arrow
    scale = lambda z: z * 2

    # Bitwise operations
    flag = a & b | (a ^ b)
    shift = a << 2 >> 1

    return flag, shift

def get_data():
    return "Simulated data"

def process(data):
    print(f"Processing: {data}")

if __name__ == "__main__":
    main()
```

Code 1. Python code example with `minted`.

You can customize its design changing `\usemintedstyle` command. The different styles offered by the `minted` package can be preview through this link — <https://pygments.org/styles/>.

8.2. Coding with Listings

Since `minted` requires additional installations and can be complex in some desktop `TeX` editors, you can use the `listings` package instead, which provides a simpler way to include code.

```
# Fira Code Demo - Python
def main():
    # Common operators
    a = 10
    b = 20
    if a < b and a != 0:
        print("a is less than b")

    # Comparisons
    x = (a >= 5) and (b <= 30)

    # Walrus operator
    if (value := get_data()) is not None:
        process(value)

    # Lambda with arrow
    scale = lambda z: z * 2

    # Bitwise operations
    flag = a & b | (a ^ b)
    shift = a << 2 >> 1

    return flag, shift

def get_data():
    return "Simulated data"

def process(data):
    print(f"Processing: {data}")

if __name__ == "__main__":
    main()
```

Code 2. Python code example with `listings`.

While `listings` is a simpler and widely supported package for code formatting, `minted` offers a more modern and powerful approach. It leverages the `Pygments` syntax highlighter to deliver superior coloring, language support, and styling options. One of its key advantages is the ability to easily switch between built-in color styles.

In contrast, `listings` requires manual setup for colors, fonts, and formatting rules. For users who prefer fine-tuned customization, the styling options for `listings` are organized in `tau.cls` file and can be modified at any time to suit individual preferences.

8.3. Inline Code*

As shown in this template, inline code has a custom style for improved visual appeal. To use it, simply add `\inlinecode{}` command and place your code inside the braces.

9. BIBLIOGRAPHY

Bibliography management is handled by `biblatex`, with the default citation and reference style set to IEEE. The `citestyle=numeric-comp` option is enabled, allowing multiple citations to be grouped within a single bracket (e.g., [1, 2]). The citation format can be customized directly in `tau.cls` to suit any preferred style.

10. FAQ

How do I manage my references?

To manage your references, I recommend using the tool `scribbr`. You can simply enter the URL or create your own citation, and then export it to `LaTeX` using the options in the three-dot menu.

The generated citation can be copied and pasted into `tau.bib`, the file designated for bibliography management. You may rename this

file, but if you do, remember to update the `\addbibresource` command in `tau.cls` under the biblatex section.

Note

Some platforms, such as Google Scholar or scientific journals, provide citations directly in BibTeX format. Therefore, check if there is a “how to cite this document” section to streamline the citation process even further.

If you have any further questions, you can refer to the following page — [Bibliography management with biblatex](#).

What should I do with the example files?

The template includes sample content — such as an example figure, bibliography entries, and the `example.py` script — to demonstrate how the layout and features work. Once you start customizing the document for your own use, feel free to delete all example files and entries you don’t need.

How do I place equations easily?

For equations, we have two options: inline or on its own line. For inline equations, simply place a dollar sign (\$) at the beginning and end of the equation. However, if you want the equation to be displayed on its own line, you need to use the equation environment.

If you find it challenging to write formulas directly in $\text{\LaTeX}$, you can use text editors like Word. In the equations menu, you can select $\text{\LaTeX}$ in the conversion section and copy and paste the equation you wrote into one of these two environments.

How do I change the paper size?

By default, this class was adapted for a4paper and test it with letterpaper. The following paper sizes are available in $\text{\LaTeX}$:

- letterpaper (11 × 8.5 in)
- legalpaper (14 × 8.5 in)
- executivepaper (10.5 × 7.25 in)
- a4paper (21 × 29.7 cm)
- a5paper (21 × 14.8 cm)
- b5paper (25 × 17.6 cm)

CONTACT ME

Have questions, suggestions, or an idea for a new feature? Found a bug or working on a project you’d like to invite me to?

Feel free to reach out — I’d be happy to help, collaborate, or fix the issue. Your feedback helps me improve my templates!

✉ memo.notess1@gmail.com
 🌐 memonotess.com
 📷 [@memo.notess](#)

GITHUB REPOSITORY

Visit the repository to access the source code, track ongoing development, report issues, and stay up to date with the latest changes.

🔗 <https://github.com/MemoJimenez/Tau-class>

SUPPORTING

Did you like this class document? Check out rho-class, made for complex research articles.

🔗 <https://es.overleaf.com/latex/templates/rho-class>

Any contributions are welcome!

Coffee keeps me awake and helps me create better $\text{\LaTeX}$ templates. If you wish to support my work, you can do so through PayPal:

👉 <https://www.paypal.me/GuillermoJimenez>

Enjoy writing with tau-class!

REFERENCES

- [1] E. Bell, C. Moss, and D. Yu, “The impact of monospaced font ligatures on code readability in scientific documentation”, *Journal of Technical Typography*, vol. 7, no. 1, pp. 33–47, 2022. DOI: [10.8888/jtt.2022.00701](https://doi.org/10.8888/jtt.2022.00701)
- [2] L. Davis and N. Kim, “Signalflow: A modular framework for real-time physiological data visualization”, *International Journal of Biomedical Computing*, vol. 15, no. 4, pp. 78–92, 2023. DOI: [10.9999/ijbc.2023.01504](https://doi.org/10.9999/ijbc.2023.01504)
- [3] PGFPlots, *A latex package to create plots*. [Online]. Available: <https://pgfplots.sourceforge.net/>